

CSE 60321 – Advanced Computer Architecture – Spring 2026

Homework 2 – 100 points

Assigned: Tuesday, February 10, 2026 – **Due:** Thursday, February 19, 2026
Individual Homework

Before you start:

1. Please clearly indicate which question an answer is for in your submission.
2. Different homework assignments may have different total point values, but they contribute equally to final grades.
3. Some problems recommend or require you to use CSE student machines to complete, which may be crowded with various minor issues at any time. In addition, programs may take longer than you expect to run. Please go through all problems before starting so that you may better organize your time in advance.
4. Unit clarification: for all problems in this homework, $1TB = 2^{10}GB$, $1GB = 2^{10}MB$, $1MB = 2^{10}KB$, $1KB = 2^{10}Bytes$, $1Byte = 8bits$; $1ns = 10^{-9}s$, $1ms = 10^{-3}s$; $1GHz = 10^9Hz$, $1Hz = 1s^{-1}$.

Problem 1: (21 points)

One possible organization of the memory hierarchy on an Intel Xeon chip is:

- A 64KB L1 instruction cache.
 - The L1 instruction cache contains only instruction encodings.
- A 64KB L1 data cache.
 - The L1 data cache contains only data words.
- A 512KB unified L2 cache.
 - The L2 cache can contain both data and instruction encodings.
- An 8 MB unified L3 cache.
 - The L3 cache can contain both data and instruction encodings.

Suppose this chip has the clock frequency at 2.0 GHz, and for its cache system, assume that:

- All caches use the block size of 128 Bytes.
- The hit time for either L1 cache is 1 clock cycle (CC); the hit time for L2 cache is 10 CCs.
- The time required to find data (i.e., hit time) in the L3 cache is 50 CCs.
- If data is not found in the L3 cache, the processor must look for data in main memory.
 - The processor-to-memory bus can accommodate 16 Bytes of memory at a time.
 - The L3 miss penalty is the initial time required to access memory, which accounts for a single transfer (equal to the bus bandwidth, in this case 16 Bytes), plus 10 CCs for every additional transfer (also 16 Bytes here).

Hint: for a cache miss, cache miss penalty (transfer time) is an additional cost on top of the hit time (access time). A cache miss happens after first attempting to access it.

Question A: (3 points)

Write “True” for correct statements, and “False” for wrong statements.

- _____ (1) This chip will be faster if we remove L2 and L3 caches and keep L1 cache the same.
- _____ (2) Typically, building a certain amount of L1 cache into the chip is more expensive than building the same amount of L3 cache.
- _____ (3) To execute a program correctly on this chip, all instructions of this program must be loaded into L1 instruction cache prior to execution.

Question B: (6 points)

Consider the L2 cache. It is 4-way set associative. Physical addresses are 64 bits long. Each data word is 64-bit. Entries are Byte addressable. **How many bits of a 64-bit physical address comprise the tag, index and offset?** Please provide your calculation process.

Question C: (10 points)

Assuming the local miss rates of 5%, 40%, and 20% for the L1, L2, and L3 caches, respectively. **To have an average memory access time of no higher than 4 CCs, determine the maximum initial main memory access time (in nanosecond).** Please provide your calculation process.

Hint: to simplify calculation, you may use CCs for time and convert to nanoseconds at the last step.

Question D: (2 points)

Assuming the same local miss rates in Question C. **Fill in the blank:**

If the bus bandwidth increases to 32 Bytes, the average memory access time will be _____ CCs.

Hint: be careful about the unit of time this question asks for.

Problem 2: (10 points)

Consider the following parameters of a virtual memory system:

Table 1: basic specifications of the system

Virtual address	Physical memory size	Page size	PTE size
30 bits	2 GB	64 KB	4 Bytes

Question A: (2 points)

Fill in the table:

Table 2: virtual address components

Page offset / bits	Virtual page memory (VPN) / bits	Total / bits
		30

Question B: (4 points)

How many page table entries (PTEs) are needed and how much physical memory is needed for storing the page table? Direct answers are expected (e.g., 2^{10} entries and 1 KB physical memory).

Question C: (4 points)

What are the pros and cons of increasing the page size? Please provide at least 1 pro and 1 con.

Problem 3: (8 points)

We are trying to deploy the way-prediction technique on an L1 instruction cache. The configurations of this cache are:

- 4-way set associative.
- 64 Bytes block size.
- 256 KB cache size.
- 1.1 ns cache access time.
- 1% cache miss rate.
- 50 ns additional cache miss penalty if missing.

With way-predicting enabled, this cache has:

- 1 ns cache access time if prediction succeeds, which is the same as a direct mapped cache.
- 1.1 ns **additional** cache access time if prediction fails.
- Other conditions remain the same.

Question A: (2 points)

Fill in the blank:

For this way-predicting enabled cache, for the best case, a cache access takes _____ ns; for the worst case, it takes _____ ns.

Question B: (6 points)

Assuming the same cache miss rate with way-predicting. The way-predicting cache reduces the AMAT by at least 5%. **What is the minimum prediction accuracy?** Please provide your calculation process.

Problem 4: (16 points)

Consider applying critical-word first and early restart techniques to reduce miss penalty in a system with L1 and L2 caches. Originally, in a cache miss scenario, the processor continues only when the entire block with the critical word is loaded into the cache.

In this problem, a word is 8 Bytes wide.

Question A: (8 points)

First, we have an L2 cache with 1MB capacity and 64-Byte block size. With critical-word first, it takes 110 CCs to get the first word from memory. Otherwise, the time to receive the first word from memory is 90 CCs, and each additional word from the main memory requires 10 CCs. Further assume that the probability of referencing the first word in a block of L2 cache is 30% while the rest ones are referenced with equal probability.

How many CCs would it take on average to service an L2 cache miss for 1) with neither critical-word first nor early restart, 2) with early restart only, and 3) with both critical-word first and early restart? Please provide your calculation process.

Question B: (4 points)

For this question, ignore Question A. The following paragraph is a brief analysis regarding L1 cache. **Fill in the blank with the most appropriate integers:**

Next, we have an L1 cache with 32 KB capacity and 32-Byte block size. To simplify the question, we only use the average L2 cache access time, i.e., we ignore how different situations of L2 cache access may affect L1 cache access. With critical-word first, it takes 27 CCs to get the first word from L2 cache. Otherwise, the time to receive the first word from L2 cache is _____ CCs, and each additional word from the main memory requires 4 CCs. Further assume that the probability of referencing every word in a block of L1 cache is the same. Then, applying both critical-word first and early restart can reduce _____% of average time for the original case without either critical-word first or early restart to service an L1 cache miss; applying only early restart can reduce 20% of average time for the original case to service an L1 cache miss.

Question C: (4 points)

Would critical-word first and early restart be more important for L1 or L2 cache? Why? What factors would contribute to their relative importance? You may refer to your observations while solving Questions A and B.

Hint: the solution is not fixed as long as it makes sense, but there must be at least one reason, and your answer and reason(s) must be consistent.

Problem 5: (45 points)

Important Notice: the following tutorial has been tested on `student04.cse.nd.edu`. Due to certain issues of operating systems, libraries, etc., Multi2Sim **DOES NOT work on other student machines**. In other words, you **have to share** student04 for this problem. This is yet another reason you should finish your homework ASAP: a lot of students will flow to these two student machines all together before due, causing performance and even connection issues.

We have already known basic usages of Multi2Sim in Homework 1. On top of that, it is possible to change the configuration of a cache and study its impact on program performance with Multi2Sim. To get started, download and extract necessary files at your home directory on a CSE student machine:

```
wget https://notredame.box.com/shared/static/nok9rzzbc9d59jbs7jgm5xb5h0rhr1tm.tar
-O hw2.tar
tar -xvf hw2.tar
```

At the beginning of `hw2/mem-config`, you can see the configurations of L1 cache:

```
head hw2/mem-config -n 7
```

Output:

```
[CacheGeometry geo-l1]
Sets = 256
Assoc = 2
BlockSize = 8
Latency = 2
Policy = LRU
Ports = 2
```

Note: you may also use `cat hw2/mem-config` or `vim hw2/mem-config` to check the content of a file.

Here, the first line is the name of the cache. Next, `Sets`, `Assoc`, `BlockSize`, and `Policy` represent number of sets, associativity, number of Bytes per block, and replacement policy, respectively. For this example, instruction and data share this L1 cache, and LRU stands for least recently used.

For this homework, you can assume that the instruction and data cache share the same configuration, and that there will always be an instruction AND a data cache. Below, we only focus on the data cache.

An understanding of the underlying processor-memory architecture can actually make you a better programmer. In this problem, we'll use matrix multiplication code as a case study. Matrix multiplication is common in scientific computing, graphics transformations, etc. There are two matrix multiplication algorithms provided: one is in `hw2/no_blocking.cc`, and the other is in `hw2/blocking.cc`. Both are multiplying two 120×120 matrices, whose original numbers are listed in `hw2/mat-mat.hh`.

Hint:

1. For usages of two programs, please refer to `hw2/README.md` or `hw2/README.pdf`.
2. To better manage configuration files of different cache settings, refer to previous tutorials for copying, renaming, and editing files remotely, or uploading files you locally create and edit. For instance, to prepare the configuration files for cache setting #1 of Question B, create a new directory (suppose you start at the home directory after extraction) and copy files:

```
cd hw2
```

```
mkdir setting-1
cp x86-config setting-1/
cp mem-config setting-1/
```

Next, use `vim setting-1/mem-config` to edit cache configurations as needed.

3. Each simulation takes approximately 15~25 minutes on student04. If you run simulations sequentially on student04, a generous allowance of about 4~6 hours for all Questions in Problem 5 is recommended.

4. You may use multiple separate terminals to connect to CSE student machines so that you can run more than one simulation at the same time. These terminals are independent as long as they are not writing to the same file(s), so please make sure that output files specified by `--x86-report` and `--mem-report` are different for each simulation.

5. Please start ASAP. All simulation time estimations above are measured on basically idle student machines. If the machines get busy (due to multiple terminal connections and/or multiple students working at the same time), the actual simulation time would be even longer.

Question A: (2 points)

Read codes of both algorithms. **What does the blocking version do differently than the more conventional “no-blocking” version?**

Question B: (15 points)

Given the following cache configurations:

Table 3: cache settings to test in Question B, Problem 5

Cache Setting	Cache Size / KB	Cache Block Size / Byte	Number of Sets	Set Associativity
Setting #1	4	8		Direct mapped
Setting #2	4	8		2-way
Setting #3	4	8		4-way

Fill in the blanks in Table 3. Then, compile two programs and run simulations for two programs under all three cache settings. For the blocking program, use two values for B: 15 and 30. **Fill in Table 4 with execution cycles:**

Table 4: execution cycles of tests in Question B, Problem 5 (Unit: CC)

Cache Setting	No-blocking	Blocking (B = 15)	Blocking (B = 30)
Setting #1			
Setting #2			
Setting #3			

Question C: (4 points)

In your results of Question B, which cache setting is the best in terms of minimum execution cycles? Give one probable reason for the performance differences among these settings.

Question D: (8 points)

For your results of Question B, calculate the speedup of two blocking results over the no-blocking result under the best cache setting you find in Question C. In this case, which blocking parameter is better? Give one probable reason for the speedup difference between different blocking parameters.

Hint: what is the equivalent B for no-blocking?

Question E: (8 points)

Starting with your best cache setting from Question C, change the cache block size to 16 Bytes and run simulations for two programs. For the blocking program, you may run with only the better B setting you determine in Question D. **To maintain the same cache size of 4 KB and the same set associativity, what is the new number of sets? Report execution cycles, compute the new speedup result, and briefly explain your findings (why execution time and speedup results are different from the results in Question D, etc.).**

Question F: (8 points)

Starting with the cache setting in Question E, change the number of sets so that the cache size becomes 8 KB, and run simulations for two programs. For the blocking program, you may run with only the better B setting you determine in Question D. **What is the new number of sets? Report execution cycles, compute the new speedup result, and briefly explain your findings (why execution time and speedup results are different from the results in Questions D and E, etc.).**